

From Industrial to Household Fans: Quantifying Domain Shift and Limited-Data Fine-Tuning in Acoustic Anomaly Detection

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ABSTRACT

Reconstruction-based acoustic anomaly detectors degrade on new hardware, but the degradation, the target data needed and the layers worth fine-tuning have not been systematically quantified. From 5 autoencoders pretrained on MIMII fans (4 single-ID, 1 mixed-ID), we study a small shift (another machine ID) and a large shift (a self-recorded household fan), fine-tuning with 5/10/20/40 normal target clips under 4 layer schemes (large shift: leave-one-round-out cross-validation). (1) The threshold-shift ratio and the AUC drop disagree across the two shifts: the small shift moves the decision threshold by only $2.42\times$ yet costs 0.347 AUC, with 6 of 12 pairs inverted, whereas the large shift moves it by $6.02\times$ but costs only 0.094. (2) Full fine-tuning reaches AUC 0.984 with 20 clips under the large shift, while 5 clips already approach the plateau under the small shift. (3) Decoder-only tuning attains 82%–97% of the full fine-tuning gain with about half the parameters; encoder-only and bottleneck-only tuning give negative gains under the large shift. (4) Pretraining helps most at 20 clips (+0.228); the advantage of mixed-ID pretraining is small and scheme-dependent, and the single-run evidence is weak. With 48 of 50 folds at AUC 1.0 ($n=40$), the recordings are highly separable, so the results are best read as a case study.

KEYWORDS

acoustic anomaly detection, domain shift, limited-data adaptation, fine-tuning, autoencoder, fan

1 INTRODUCTION

Listening to a machine with a microphone is a low-cost, non-invasive way to monitor its condition. The MIMII dataset [1] established the standard recipe for this task: train an autoencoder on normal sounds only and use the reconstruction error as the anomaly score. Its training data, however, come from a research-grade eight-channel microphone array in a controlled environment, whereas a real deployment is often a single consumer microphone on a different machine. Acoustic domain shift has therefore become a central topic in the DCASE community: DCASE 2021 Task 2 made “acoustic characteristics differ between training and test samples” a task condition [2], and the survey by Wilkinghoff et al. points out that changing the microphone type or sensor position can affect the signal more than the anomaly itself, while users are usually willing to provide only very few target-domain samples [3].

Prior work mostly studies shift within a single recording infrastructure. We ask two questions closer to deployment: what happens when a model trained on an industrial fan is moved to another machine ID, and to a completely different consumer household fan;

and, when only a few to a few dozen target-domain normal clips are available, how to fine-tune most cost-effectively. We do not propose a new method; we carry out a systematic quantification that is as clean as we can make it.

Specifically, we answer three research questions:

- **RQ1 (quantifying domain shift):** How does the degradation caused by switching machine IDs within a product line (small shift) differ in nature from that caused by moving across hardware (large shift)?
- **RQ2 (data efficiency):** How many target-domain normal clips are needed for effective adaptation, and does this depend on the size of the shift?
- **RQ3 (fine-tuning strategy):** Which layers are most effective to tune? How do target-domain gain and source-domain forgetting trade off? Does seeing more machine IDs during pretraining reduce the data requirement?

The experiments comprise 5 pretrained models, frozen evaluation on 12 small-shift pairs and 5 large-shift models, 272 fine-tuning configurations (992 training runs), and 4 no-pretraining control configurations (40 training runs): 1037 training runs in total, with 0 failures. Our contributions are:

- (1) We find that the threshold-shift ratio and the AUC drop are inconsistent across the two shifts, and that the small shift inverts the normal/abnormal ranking (Section 4.1).
- (2) We give data-efficiency curves for both shifts: under the large shift the gain slows markedly after 20 clips, and under the small shift 5 clips already approach the plateau (Section 4.2).
- (3) We systematically compare four layer schemes and explain their success or failure with layer-wise weight drift: the decoder is key, one-sided tuning hurts across hardware, and bottleneck-only tuning is harmful on both counts (Section 4.3).
- (4) We quantify the value of pretraining itself, test the role of pretraining diversity (its advantage is small and depends on the layer scheme), and report the evidence that the results at $n=40$ are “too perfect” (Sections 4.3.4, 4.4 and 6).

2 RELATED WORK

The MIMII dataset and baseline. MIMII [1] contains normal and abnormal sounds of 7 machine IDs for each of four machine types (valves, pumps, fans and slide rails), recorded as 10-second clips at 16 kHz with an eight-channel TAMAGO-03 microphone array and mixed with real factory noise at three signal-to-noise ratios (we use the 0 dB version). Its baseline uses only the first channel, computes a log-Mel spectrogram with 64 Mel filters, a frame length of 1024 and a hop of 512, concatenates 5 adjacent frames

into a 320-dimensional input vector, trains a 320–64–64–8–64–64–320 fully connected autoencoder (ReLU hidden layers, no activation on the output layer) with Adam for 50 epochs, flags anomalies by the mean reconstruction error over the 10-second clip, and evaluates with AUC; the test set consists of all abnormal clips plus an equal number of randomly drawn normal clips, and the remaining normal clips are used for training. Our features, architecture and pretraining recipe follow this baseline.

Domain shift in DCASE. DCASE 2021 Task 2 [2] introduced domain-shift conditions into unsupervised anomalous sound detection: the acoustic characteristics of training and test samples differ because of seasons, products or environmental noise. Wilkinghoff et al. [3] review the domain-generalization problem in DCASE-related work, noting that a model must be sensitive to subtle anomalies yet insensitive to domain changes such as microphone type and sensor position, and that the target domain can usually be characterized by only very few target-domain samples.

Positioning. The work above defines shift within a single recording infrastructure. We split shift into two levels, “switching machine IDs” and “switching hardware, environment and microphone”, and quantify the amount of fine-tuning data, the layer scheme and the cost of forgetting within one set of experiments.

3 EXPERIMENTAL SETUP

3.1 Source Domain: MIMII Fans

We use four machine IDs from the 0 dB version of the MIMII fan subset: id_00 (1011 normal and 407 abnormal clips), id_02 (1016 / 359), id_04 (1033 / 348) and id_06 (1015 / 361). For each machine ID we first draw 160 normal + 320 abnormal clips as its fixed evaluation set (normal:abnormal = 1:2, matching the target domain; see Section 3.6 for the reason); the remaining normal clips form the training pool: id_00 851, id_02 856, id_04 873 and id_06 855 clips. Each of the four single-ID models is trained on its own training pool. The mixed model merged draws 215 clips from each machine ID’s training pool, 860 clips in total, and its evaluation set takes 40 normal + 80 abnormal clips from each machine ID’s fixed evaluation set, giving 160 + 320. All sampling uses the fixed random seed 42, and training pools never overlap with evaluation sets; when fine-tuning under the small shift, target-ID samples are drawn only from that ID’s training pool and never from its evaluation set.

3.2 Target Domain: A Self-Recorded Household Fan

The target domain consists of 180 self-recorded clips of a Wathai 120 mm AC axial fan (Figure 1): 3 operating conditions $\times$ 3 voltages (4 V / 8 V / 12 V) $\times$ 2 environments (quiet / noise) $\times$ 10 rounds, each clip 10 seconds long, 16 kHz and mono, recorded with a Razer Seiren V3 Mini USB microphone. The three operating conditions are: normal (healthy operation); blocked (a board placed in front of the fan partially obstructs the intake airflow); and imbalance (an extra weight attached to one blade simulates rotational imbalance). In the noise environment, a speaker next to the fan plays a recording of an operating air conditioner as background noise; in the quiet environment nothing is played. Condition, voltage, environment and round are encoded in the file name {condition}_{voltage}_{noise}_run{XX}.wav. The 60 normal

clips are used for fine-tuning and training; the 120 blocked + imbalance clips are used only for evaluation and never enter any training step (the autoencoder learns only normal sounds).

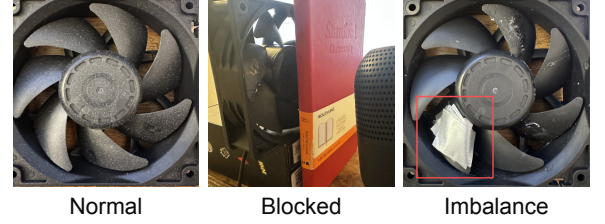


Figure 1: The three operating conditions of the target fan. Blocked: a hardcover notebook stands in front of the fan as a baffle, next to the speaker that plays the air-conditioner noise. **Imbalance:** the red box marks the weight taped to a blade.

3.3 Features and Model

Feature extraction follows the MIMII baseline: audio is read as mono at its original sampling rate, a Mel spectrogram is computed with 64 Mel filters, a frame length of 1024 and a hop of 512, converted as $S_{\log} = (20/p) \log_{10}(S + \epsilon)$ ($p = 2.0$; the formula follows the MIMII baseline [1]), and 5 adjacent frames are concatenated into a 320-dimensional vector; each 10-second clip yields 309 vectors. The model is a 320–64–64–8–64–64–320 fully connected autoencoder (Figure 2) with ReLU hidden layers and no activation on the output layer, 50,760 parameters in total. The anomaly score of a clip is the mean, over all its vectors, of the reconstruction mean squared error:

$$\text{Score}(f) = \frac{1}{T} \sum_{t=1}^T \frac{1}{320} \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|^2, \quad T \approx 309. \quad (1)$$

3.4 Four Stages

Stage 0: pretraining. We train 5 models with the recipe above (Adam, learning rate 10^{-3} , batch size 512, 50 epochs, 10% of the samples held out for validation, used only to monitor the loss). Right after training, AUC and best F1 are recorded on each model’s 480-clip source evaluation set as the “source baseline”.

Stage 1: domain-shift quantification (frozen evaluation). The parameters of all 5 models are kept fixed. Small shift: each of the 4 single-ID models is evaluated on the other 3 machine IDs it has never seen (using the target ID’s 160 + 320 evaluation set), giving 12 pairs; merged has seen all four machine IDs and is excluded from the small shift. Large shift: all 5 models are evaluated on the self-recorded fan and aggregated over 10 folds defined by the leave-one-round-out cross-validation of Section 3.5 (exactly the same split as in Stage 2), so that the numbers of the two stages can be subtracted directly.

Stage 2: main fine-tuning experiment. The grid is data level $\times$ layer scheme. The data levels are 5 / 10 / 20 / 40 target-domain normal clips, drawn in a nested way (clips are first ordered with a fixed seed and level n takes the first n , so $5 \subset 10 \subset 20 \subset 40$). There

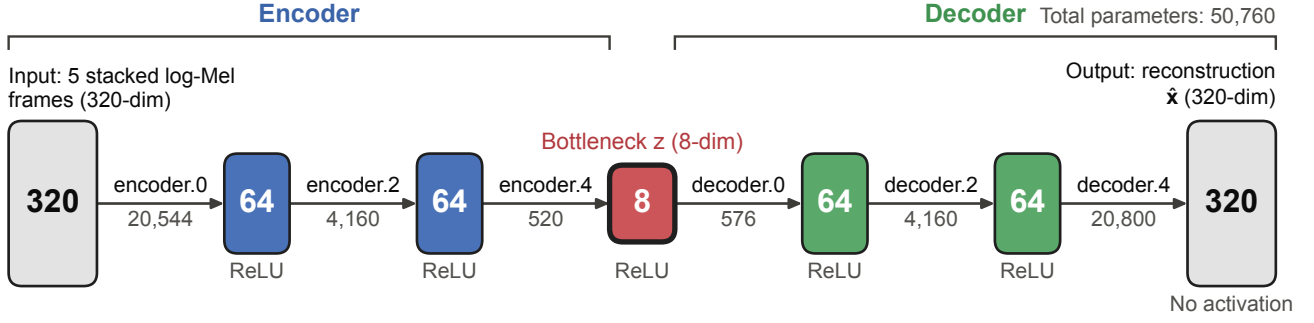


Figure 2: Autoencoder architecture. Above each arrow is the name of the linear layer; below it is that layer’s parameter count.

are four layer schemes: full fine-tuning, decoder-only, encoder-only and bottleneck-only (the two layers into and out of the 8-dimensional bottleneck, encoder.4 and decoder.0; see Figure 2 for the layer names). The hyperparameters are the same throughout: Adam, learning rate 10^{-4} , 30 epochs, batch size 64, no early stopping and no source-data replay. Small shift: 4 starting models $\times$ 3 targets $\times$ 4 levels $\times$ 4 schemes = 192 configurations, each trained once. Large shift: 5 starting models $\times$ 4 levels $\times$ 4 schemes = 80 configurations, each run over 10 folds and restarted from the pretrained weights in every fold, 800 runs in total. Altogether this gives 272 configurations and 992 training runs (Table 1). Five clips contain about 1545 vectors, giving 750 gradient steps over 30 epochs; 40 clips give 12,360 vectors and 5820 steps.

Stage 3: no-pretraining control. The same architecture is randomly initialized and the whole network is trained on target-domain normal clips only, with the same hyperparameters as in Stage 2. A randomly initialized model has no “learned encoder or decoder” to speak of, so there is only one training mode: 4 data levels $\times$ 10 folds = 40 training runs.

Table 1: Scale of the experiments. Stage 1 evaluates 12 small-shift pairs and 5 large-shift models; configs = configurations.

Stage	Content	Runs
0 Pretraining	4 single-ID + 1 mixed	5
1 Frozen evaluation	12 small + 5 large	0
2 Fine-tuning	272 configs: 192 + 80 $\times$ 10 folds	992
3 No pretraining	4 levels $\times$ 10 folds	40
Total	0 failures	1037

3.5 Leave-One-Round-Out Cross-Validation (LORO)

The 180 clips consist of 18 condition combinations, each recorded in 10 rounds. The 10 repetitions of a condition were captured consecutively on the same device, at the same position and within a short time, so they are highly similar; a random split would place different rounds of the same condition on both the training and test sides, and would measure memorization rather than generalization. The large shift therefore uses leave-one-round-out cross-validation: fold k ($k = 0, \dots, 9$) holds out the 18 clips of round $k + 1$

(6 normal + 12 abnormal) as the test set, leaving 54 normal clips available for training, which is why the data levels stop at 40 clips; every fold is fine-tuned again from the pretrained weights rather than continuing from the previous fold’s model. The 10 folds are aggregated in two ways: *pooled*, where the scores of the test samples of all 10 folds are concatenated into 180 clips and one AUC is computed (the main number); and *per-fold*, where each fold is computed separately and the mean and standard deviation are reported. The small-shift evaluation set is the target ID’s 160 + 320 clips; MIMII has no structure of consecutive recordings of the same condition, so LORO is not used there.

3.6 Metrics

AUC is the primary metric: the probability that, for a randomly drawn abnormal sample and a randomly drawn normal sample, the abnormal one has the higher score, $AUC = P(s_{\text{abnormal}} > s_{\text{normal}})$. It measures the ability to rank anomalies above normal samples and does not depend on the threshold (0.5 is chance). **Best F1** is secondary: $F_1 = 2PR/(P + R)$, where $P = TP/(TP + FP)$ and $R = TP/(TP + FN)$; with abnormal as the positive class, TP, FP and FN are the numbers of abnormal samples flagged as abnormal, normal samples flagged as abnormal, and abnormal samples flagged as normal. We scan 1000 equally spaced thresholds over the score range, flag “score above threshold” as abnormal, and take the maximum F1. Best F1 has a trivial floor set by the class ratio: with normal:abnormal = 1:2, flagging every sample as abnormal already gives $F1 = 0.800$. We therefore also set the source evaluation sets to 1:2 so that the floors match on both sides, but all conclusions are stated in terms of AUC.

The following four metrics are defined by us for this analysis. **Gain** = fine-tuned target AUC – frozen target AUC from Stage 1 (pooled for the large shift; the same fold is used for fold-wise comparisons). **Forgetting** = Stage 0 source AUC – AUC re-measured on the same source evaluation set after fine-tuning (the 10-fold mean for the large shift). **Threshold-shift ratio** = median reconstruction error of target-domain normal samples / median reconstruction error of the normal samples in the source evaluation set; the numerator uses the 160 normal clips of the target ID’s evaluation set for the small shift and all 60 normal clips for the large shift (frozen evaluation has no leakage risk, and 60 clips are more stable than the 6 of a single fold). **Weight drift** $\|\Delta W\|/\|W\|$ is, for each

linear layer, the Frobenius norm of the change of its weight matrix during fine-tuning divided by its norm before fine-tuning; frozen layers are always 0.

4 RESULTS

4.1 RQ1: The Two Shifts Degrade the Model Differently

Finding 1: Threshold-shift ratio and AUC drop disagree. Under the small shift the threshold-shift ratio averages 2.42 (range 1.38–3.99), yet AUC drops by 0.347 on average (range 0.135–0.610); under the large shift the threshold shifts by $6.02\times$ (4.60–8.36), yet AUC drops by only 0.094 (Figure 3). Threshold shift reflects a drift in the scale of the reconstruction error and can be handled by recalibrating a single number; the AUC drop reflects a loss of ranking ability and cannot be repaired by calibration. The two are inconsistent in our experiments, which shows that the “size of a shift” cannot be captured by a single metric.

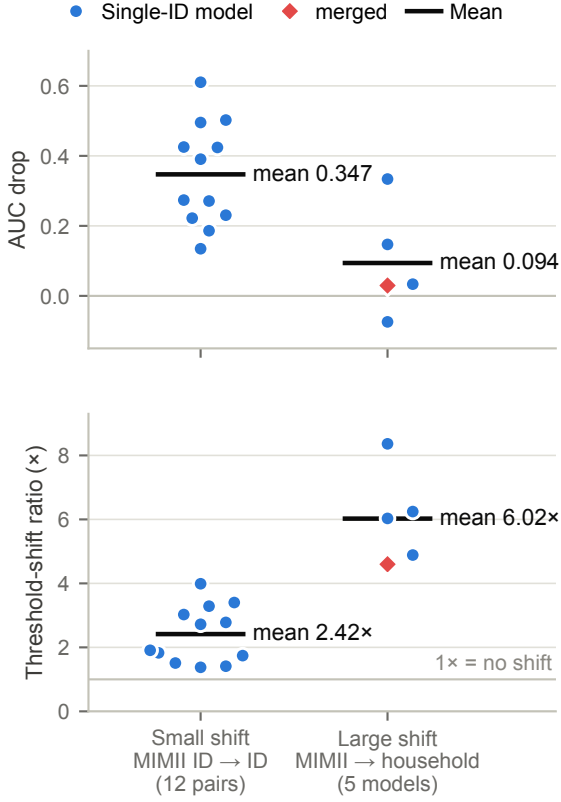


Figure 3: Frozen evaluation under the two domain shifts: AUC drop (top; source AUC – target AUC) and threshold-shift ratio (bottom).

Finding 2: The small shift does not blunt the model; it inverts the ranking. In 6 of the 12 pairs the frozen target AUC is below 0.5, with a minimum of 0.257 (id_00 → id_06); the mean

over the 12 pairs is only 0.479 and the maximum only 0.655 (Figure 4, top). An AUC below 0.5 means the model ranks faulty samples as more “normal” than normal ones. By target ID, id_06 is the hardest: the three models that evaluate it reach a mean AUC of 0.300. By starting model, id_06, which has the highest source AUC (0.984), also suffers the largest drop when moved to other IDs (mean 0.475). One possible explanation is that machine IDs within a product line differ only subtly, so the “normal template” learned by a model fits its own ID very closely, and after switching IDs the normal patterns of the original ID end up closer to the anomalies of the new one; this is speculation that our experiments cannot verify directly.

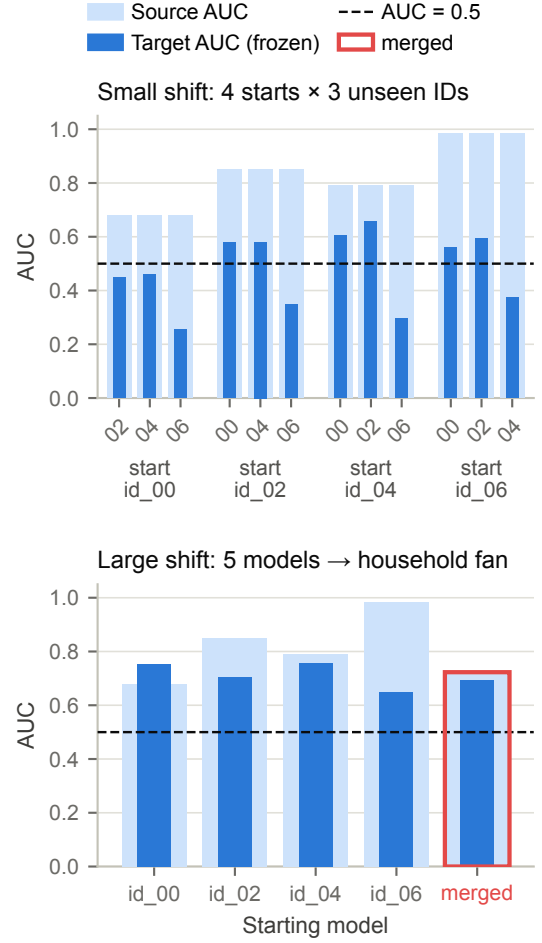


Figure 4: Source AUC and frozen target AUC. In the top panel, the ticks within each group are target machine IDs (02 stands for id_02).

Finding 3: The large shift only blunts the model; it does not invert the ranking. The frozen pooled AUC of the 5 models on the self-recorded fan lies between 0.650 and 0.756, all above 0.5 (Figure 4, bottom). id_00 even improves instead of degrading (source 0.681, target 0.755, AUC drop -0.074); the largest drop again belongs to id_06, the strongest source model (0.984 →

0.650, a drop of 0.334). The frozen AUC drop of merged is 0.030 with a threshold shift of 4.60 $\times$: its threshold shift is the smallest of the 5 models, but its AUC drop is not the smallest (that of `id_00` is negative).

4.2 RQ2: About 20 Clips for the Large Shift, 5 for the Small Shift

Finding 4: For the large shift, the turning point is at 20 clips. The pooled AUC of full fine-tuning is 0.749 at 5 clips, 0.884 at 10, 0.984 at 20 and 0.998 at 40, with increments of +0.135, +0.100 and +0.014 (Figure 5, bottom). The first 20 clips (about 200 seconds of audio) deliver almost all of the gain. Two caveats apply: at 20 clips the AUC is already 0.984, so the “slowdown” from 20 to 40 clips partly reflects a ceiling effect; and with only four data levels, the turning point can only be located between 10 and 20 clips.

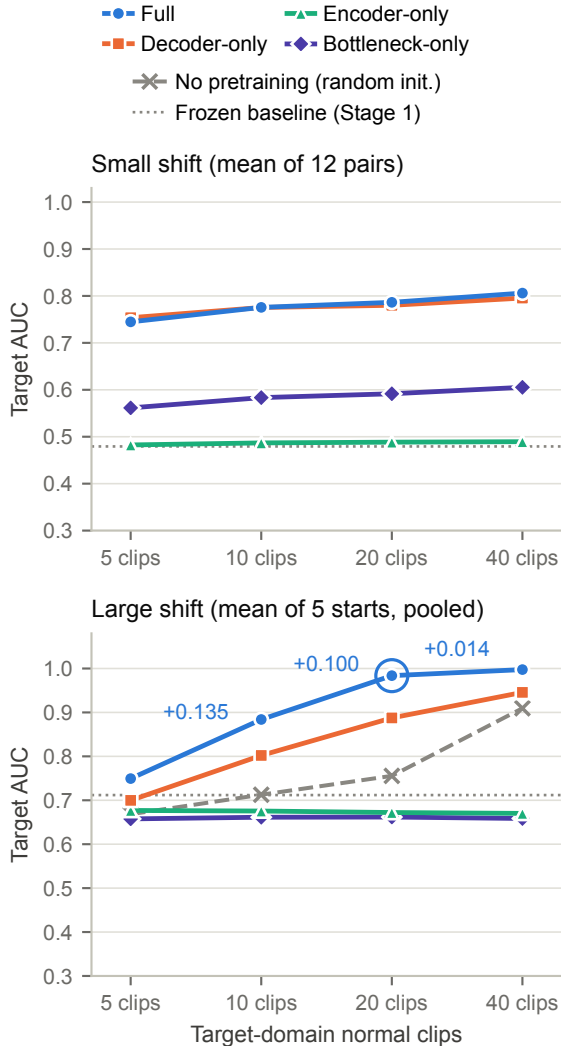


Figure 5: Target AUC as a function of the amount of fine-tuning data.

Finding 5: For the small shift, 5 clips already approach the plateau. Under the small shift, full fine-tuning reaches 0.745 at 5 clips and 0.806 at 40 clips, only 0.061 higher (Figure 5, top); encoder-only tuning barely moves across the four levels (0.482–0.489). For the small shift the bottleneck is not the amount of data but how the model is tuned (Section 4.3).

4.3 RQ3: Fine-Tuning Strategy

4.3.1 Which Layers to Tune. Finding 6: The decoder is key; one-sided tuning hurts across hardware. At $n=40$ (Table 2), full fine-tuning always gives the highest target-domain gain (small shift +0.327, large shift +0.286); decoder-only tuning, with about half of the parameters (25,536, 50%), achieves 97% (small shift, 0.316/0.327) to 82% (large shift, 0.234/0.286) of that gain. Encoder-only tuning is almost ineffective under the small shift (+0.010) and negative under the large shift (−0.042); bottleneck-only tuning gains +0.126 under the small shift but is likewise negative under the large shift (−0.053). Adapting across hardware seems to require the encoder and the decoder to be adjusted together, and moving only one side breaks their existing match; see Section 4.3.3 for the mechanism.

Table 2: Target-domain gain and source-domain forgetting of the four layer schemes at $n=40$, with their numbers of trainable parameters (small shift: mean over 12 pairs; large shift: mean over 5 starting models, pooled; forgetting under the large shift is the 10-fold mean).

Scheme	Params	Gain		Forgetting	
		Small	Large	Small	Large
Full	50,760 (100%)	+0.327	+0.286	0.346	0.367
Decoder-only	25,536 (50%)	+0.316	+0.234	0.335	0.282
Encoder-only	25,224 (50%)	+0.010	−0.042	0.023	0.089
Bottleneck-only	1,096 (2%)	+0.126	−0.053	0.065	0.249

4.3.2 Gain versus Forgetting. We describe Figure 6 with an explicit threshold: we define “forgetting < 0.15 and gain > 0.30” as the low-cost, high-gain region. Under the small shift, 20 of the 192 configurations fall into this region; under the large shift, 0 of the 80 do. These 20 configurations have something in common: their starting model is always `id_00` (with all three targets `id_02`, `id_04` and `id_06` represented); by layer scheme there are 9 full fine-tuning, 8 decoder-only and 3 bottleneck-only configurations, and the four data levels contribute 4, 5, 5 and 6; their largest forgetting is 0.122 and their smallest gain is 0.304. `id_00` has the lowest source AUC of the four single-ID models (0.681); a natural guess is that it simply has less to lose, which keeps its forgetting low. This does not indicate a free lunch that holds in general.

On average (Table 2), the forgetting of full fine-tuning is around 0.35 under both shifts (0.346 / 0.367); decoder-only tuning is of the same order (0.335 / 0.282); encoder-only tuning barely forgets (0.023 / 0.089) but also barely gains; and bottleneck-only tuning is harmful on both counts under the large shift, with a gain of −0.053 and yet a forgetting of 0.249. Apart from the cluster of `id_00` starting models, gain and forgetting are tied together in our setting (no source-data replay).

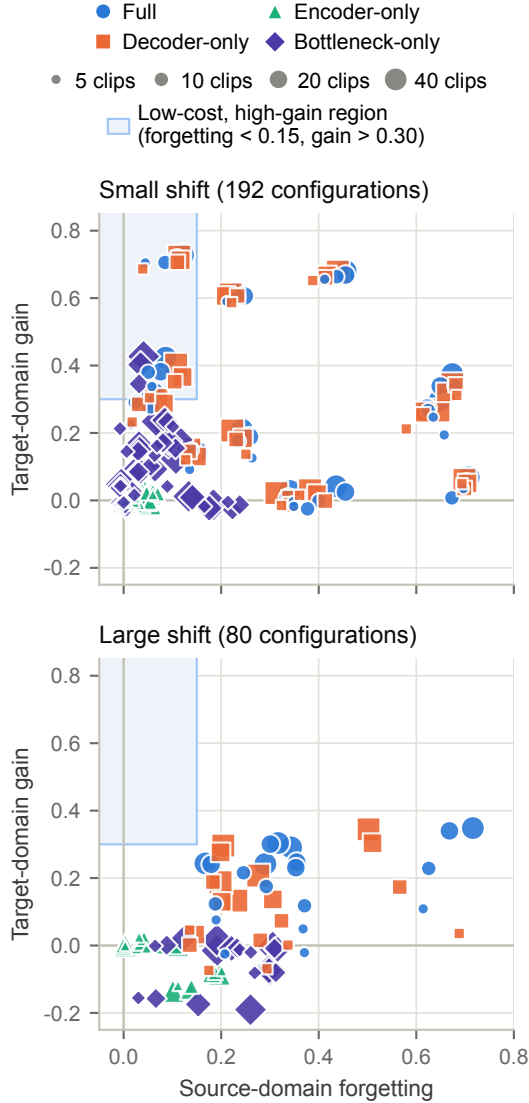


Figure 6: Target-domain gain versus source-domain forgetting for the 272 fine-tuning configurations.

4.3.3 Where the Weights Actually Moved. Appendix Figure 8 explains Finding 6. Under full fine-tuning, the layer that moves most in the small shift is the output layer decoder .4 (0.171), followed by the input layer encoder .0 (0.149); in the large shift it is the other way round: the input layer encoder .0 moves most (0.345), followed by decoder .4 (0.314) and decoder .2 (0.256). Hence “adaptation happens mainly in the decoder” holds only for the small shift; across hardware, the input and output layers at both ends change substantially while the middle layers change less. One possible explanation is that across hardware (with both the microphone and the environment changing) the distribution of the input features itself changes, so the first layer has to be re-fitted; this is only a speculation consistent with the data.

Under encoder-only tuning the encoder moves (encoder .0 0.188, encoder .4 0.189, large shift) while the decoder is frozen: the latent representation changes, but the decoder still reconstructs according to the old one, and the large-shift AUC decreases instead. Bottleneck-only tuning moves encoder .4 by 0.434, the largest single-layer change in the whole table, and decoder .0 by 0.239, while the other four layers stay fixed: the information-dense 8-dimensional bottleneck is heavily rewritten with no cooperation from the layers upstream or downstream, so the model neither learns the new domain nor keeps the old one. Fewer parameters do not mean smaller changes.

4.3.4 Pretraining Diversity: Does the Mixed Model Help? merged has no advantage under frozen evaluation: its large-shift pooled AUC is 0.693, below the mean of the 4 single-ID models, 0.716. After fine-tuning, its lead depends on the layer scheme (Figure 7). Under full fine-tuning, merged leads by +0.025 at 5 clips, +0.031 at 10 and +0.013 at 20, and the lead vanishes at 40 (0.000); that is, pretraining diversity is worth about 0.03 AUC only with the least data. This lead is of the same order as the per-fold standard deviation at the same levels (0.055 at 5 clips, 0.080 at 10 clips), so single runs cannot rule out chance. Under decoder-only tuning the lead is larger (+0.049, +0.099, +0.103, +0.055). Under encoder-only and bottleneck-only tuning the lead is about 0.04–0.06, but it must be read with care: these two schemes change the model very little, and the single-ID mean is pulled down by id_06 (at n=40, id_06 reaches 0.515 under encoder-only and 0.460 under bottleneck-only tuning). Looking only at the other three single-ID models, their mean (0.711 / 0.708) is in fact slightly higher than merged (0.703 / 0.710). The impression that “diversity lasts longer under restricted layer schemes” is therefore largely caused by the collapse of one single-ID starting model and should not be taken as a conclusion.

4.4 The Value of Pretraining Itself

We compare the randomly initialized model trained end to end on target-domain normal clips only (Stage 3) with “MIMII pretraining + full fine-tuning” (Stage 2, mean over 5 starting models) at the same data levels (Figure 5, bottom). Without pretraining the pooled AUC is 0.668 at 5 clips, 0.713 at 10, 0.756 at 20 and 0.909 at 40; with pretraining it is 0.749, 0.884, 0.984 and 0.998; the differences are +0.081, +0.171, +0.228 and +0.088, largest at 20 clips, in an inverted-U shape. The narrowing on the right is mainly a ceiling effect: the pretrained side has already reached 0.998 and cannot pull further ahead. A safer statement is that pretraining reduces the target-domain data needed to “reach 0.98 or above” from more than 40 clips to 20 clips.

5 NOTABLE OBSERVATIONS

- (1) **Threshold shift and AUC drop disagree.** The large shift, which moves the threshold most, preserves the ranking ability best (Section 4.1). The threshold is a calibration problem; AUC is a capability problem.
- (2) **Switching machine IDs inverts the ranking.** In 6 of the 12 small-shift pairs, $AUC < 0.5$. For deployment this means that when switching to another machine ID of the same product line, recalibrating the threshold is not enough; at least the decoder has to be retrained.

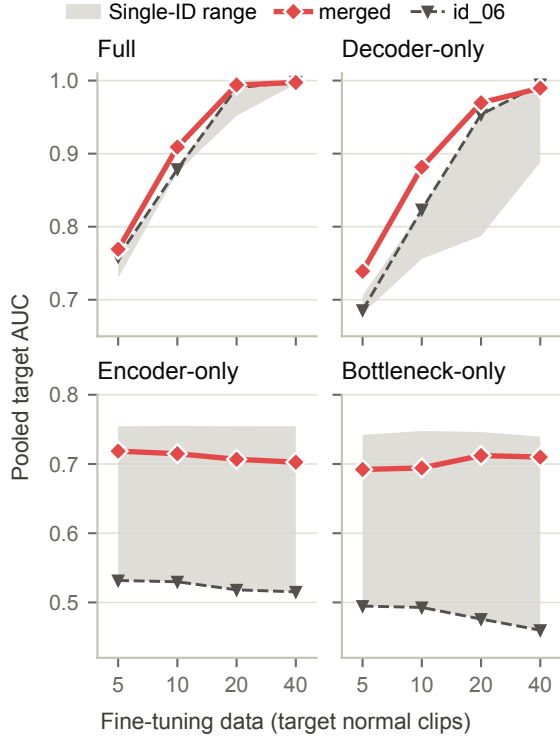


Figure 7: Pooled AUC of merged (pretrained on the mixture of four machine IDs) and the 4 single-ID starting models under the large shift. The grey band spans the minimum to maximum of the single-ID starting models; the two rows use different vertical ranges.

- (3) **Bottleneck-only tuning is harmful on both counts.** Tuning only 2% of the parameters does not mean changing the model little: the input layer of the 8-dimensional bottleneck moves by 0.434, the largest in the whole table, while the large-shift gain is negative and the forgetting is 0.249.
- (4) **id_06 is the outlier among the four IDs.** It is the hardest target (frozen mean 0.300), the starting model with the largest AUC drop and forgetting, and the target that gains the most after fine-tuning (+0.673, 0.300 $\rightarrow$ 0.973).
- (5) **The low-cost, high-gain region comes entirely from id_00.** All 20 configurations, without exception (Section 4.3.2).

6 LIMITATIONS

(1) **The results at $n=40$ are too perfect.** Under the large shift with full fine-tuning at $n=40$, of the 5 starting models $\times$ 10 folds = 50 folds, 48 reach an AUC of exactly 1.000 and the other two reach 0.986, and four of the starting models have a per-fold standard deviation of 0 (Appendix Figure 9, top, and Appendix Table 3). LORO prevents the same round from appearing on both the training and test sides, but the 180 clips are 10 rounds recorded consecutively

on the same day, with the same device, at the same position, so different rounds may still be overly similar: the environmental noise, microphone position and the fan’s condition on that day are essentially unchanged across the 10 rounds. What the model learns may be “the sound of this fan today” rather than “the sound of this kind of fan”.

(2) **Random initialization reaches 0.668 with 5 clips.** A model with no prior knowledge that has seen only 5 normal clips (about 1545 vectors) reaches a pooled AUC of 0.668 and a per-fold mean of 0.661, far above 0.5, and reaches 0.909 at 40 clips (Appendix Figure 9, bottom). This shows that blocked / imbalance differ acoustically from the normal condition by a wide margin, making this a relatively easy detection task, and further supports concern (1). The “value of pretraining” in Section 4.4 should be read against this background.

(3) **Other limitations.** The target domain has only one fan and one microphone, so all large-shift conclusions are a case study. All experiments are single runs with seed 42, and the lead of merged in Section 4.3.4 is within the noise of a single run. The number of epochs is fixed at 30, so the 40-clip level receives 7.8 times as many gradient steps as the 5-clip level (5820 versus 750), and “more data” is confounded with “longer training” in the data-efficiency curves. Best F1 has a trivial floor of 0.800, so the text uses only AUC. The pooled AUC is not normalized per fold; the 10 folds are 10 independent models whose score scales may differ. There are only four data levels. The region thresholds in Section 4.3.2 were set post hoc for description and are not a pre-registered hypothesis.

7 CONCLUSION

Domain shift. The size of a shift cannot be captured by a single metric: the threshold-shift ratio measures the drift in score scale, the AUC drop measures the loss of ranking ability, and the two are inconsistent in our experiments. Switching machine IDs within a product line inverts the ranking, which is more damaging than the “blunting” caused by moving across hardware.

Limited-data adaptation. Adapting across hardware needs only about 20 clips (200 seconds) of target-domain normal audio to come close to perfect, and the first 20 clips deliver almost all of the gain; pretraining reduces the data needed to reach an AUC of 0.98 from more than 40 clips to 20. Switching machine IDs within a product line needs only 5 clips, provided the right layers are tuned. These numbers rest on a highly separable target dataset and should be revised upward for harder scenarios.

Fine-tuning strategy. Decoder-only tuning uses about half of the parameters and achieves 82%–97% of the gain of full fine-tuning; encoder-only tuning is almost ineffective, and bottleneck-only tuning is harmful across hardware. Across hardware, the input and output layers at both ends need substantial adjustment, and one-sided tuning breaks the existing match. Without source-data replay, gain and forgetting are broadly tied. The advantage of having seen more machine IDs during pretraining (merged) is small and depends on the layer scheme: under full fine-tuning it leads by about 0.03 only with the least data (≤ 10 clips) and vanishes as data grows; under decoder-only tuning the lead is larger; under encoder-only and bottleneck-only tuning the apparent lead

comes mainly from the collapse of the id_06 starting model. All of these are single-run results, and the evidence is weak.

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A SUPPLEMENTARY TABLE

Table 3: Per-fold AUC standard deviation under the large shift (10 folds). Top: full fine-tuning, by starting model. Bottom: by layer scheme (mean over 5 starting models).

Starting model	5 clips	10 clips	20 clips	40 clips
id_00	0.0596	0.0768	0.0042	0.0000
id_02	0.0423	0.0640	0.0362	0.0056
id_04	0.0520	0.0872	0.0255	0.0000
id_06	0.0624	0.0869	0.0167	0.0000
merged	0.0609	0.0826	0.0064	0.0000

Layer scheme	5 clips	10 clips	20 clips	40 clips
Full	0.0554	0.0795	0.0178	0.0011
Decoder-only	0.0641	0.0585	0.0415	0.0304
Encoder-only	0.0205	0.0201	0.0242	0.0234
Bottleneck-only	0.0214	0.0200	0.0186	0.0220

B SUPPLEMENTARY FIGURES

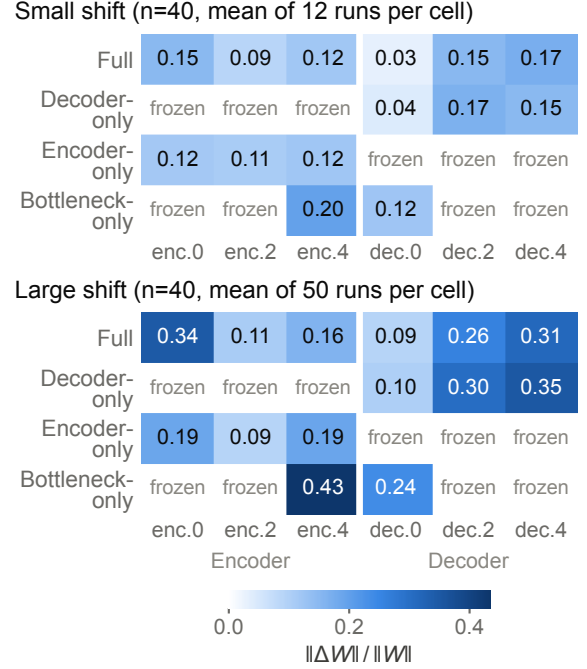


Figure 8: Relative weight drift $\|\Delta W\|/\|W\|$ of the six linear layers at n=40 (“frozen”: not trained). Dimensions: enc.0 320 → 64, enc.2 64 → 64, enc.4 64 → 8, dec.0 8 → 64, dec.2 64 → 64, dec.4 64 → 320.

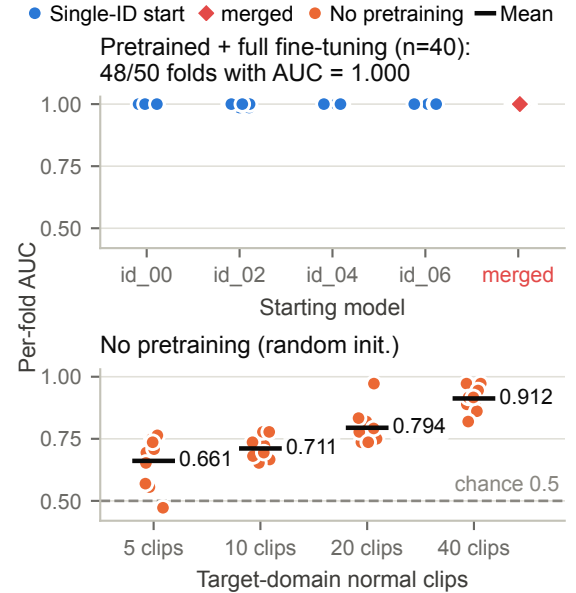


Figure 9: Per-fold AUC under the large shift (each fold: 6 normal + 12 abnormal clips); 10 folds for each starting model (top) or data level (bottom).